

★ WEEK 14 · PART 3 · PRACTICE

HMC and NUTS, Worked by Hand

Roll the Ball One Step, Then Let NUTS Stop It

The Hamiltonian, a leapfrog step, and the U-turn

Part 3's physics-powered sampler, with the arithmetic shown. We build the potential from the posterior, add momentum, and take one leapfrog step - watching the energy stay nearly conserved - then follow the ball as it swings and see NUTS end the roll at the U-turn.

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PART 3 · APPLICATIONS

Roll a Ball, Don't Wander

Case A: the Hamiltonian and one leapfrog step - potential plus kinetic energy, and how leapfrog keeps it nearly conserved so the move is almost always accepted. Case B: how far to roll, and how NUTS stops at the U-turn. Then a Claude Code lab.

What Part 3 Told Us

Now we roll the ball - with numbers

1

Random walks waste effort

Metropolis wanders aimlessly. HMC instead rolls a ball down the posterior's landscape, moving far in the right direction.

2

Energy guides the motion

The potential is the negative log posterior; momentum is a random push. Their sum, the Hamiltonian, barely changes as the ball rolls.

3

NUTS sets the one dial for you

How far to roll is the only knob HMC can't set. NUTS rolls until the ball turns back on itself, then stops.

Everything here is a calculator exercise: one leapfrog step, one energy check, one U-turn.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a position and a momentum.
- Roll with leapfrog, or follow the swing.
- Then read the energy or the U-turn.

What you should watch

- How the gradient steers the ball.
- How the total energy stays nearly flat.
- How the ball turns back on itself.

One Roll of the Ball

The Hamiltonian, and a leapfrog step

HMC turns the posterior into a landscape: the potential energy is its negative log, so high-probability regions are valleys. Add a random momentum, and the ball rolls. One leapfrog step - half-kick, drift, half-kick - moves it, and the total energy barely changes, so the move is almost always accepted. We take that step by hand

The setup

- A standardized Gaussian posterior.
- Potential $U = \text{half } \theta^2$.
- Start at $\theta = 1$, momentum = 1.

What we'll do by hand

- Take one leapfrog step, size 0.3.
- Check the total energy before and after.
- Compute the acceptance probability.

Potential, Gradient, Hamiltonian

The landscape the ball rolls on

$$\underbrace{U(\theta) = -\log p(\theta \mid D)}_{\text{the landscape}}, \quad \nabla U = \theta, \quad \underbrace{H = U(\theta) + \frac{1}{2}p^2}_{\text{potential} + \text{kinetic}}$$

The potential

- It's the negative log posterior.
- Valleys are high-probability regions.
- Its gradient is the slope the ball feels.

The Hamiltonian

- Potential plus kinetic energy.
- The kinetic part is half momentum-squared.
- It stays nearly constant as the ball rolls.

One Leapfrog Step

Half-kick, drift, half-kick

$$\text{half-kick: } p \leftarrow p - \frac{\epsilon}{2} \nabla U = 1.0 - 0.15 = 0.85$$

$$\text{drift: } \theta \leftarrow \theta + \epsilon p = 1.0 + 0.3(0.85) = 1.255$$

$$\text{half-kick: } p \leftarrow 0.85 - 0.15(1.255) = 0.662$$

The momentum gets a half-kick from the gradient, the position drifts with that momentum, then a final half-kick completes the step - moving from $\theta=1.0$ to 1.255. The gradient did the steering.

Energy Nearly Conserved

A tiny error, a tiny correction

$$H : 1.000 \rightarrow 1.0065 \ (\Delta H = +0.0065) \Rightarrow \text{accept} = \min(1, e^{-\Delta H}) = \mathbf{0.994} \text{ (energy nearly conserved)}$$

The check

- The Hamiltonian went from 1.000 to 1.0065.
- A change of just 0.0065.
- Leapfrog keeps energy almost exact.

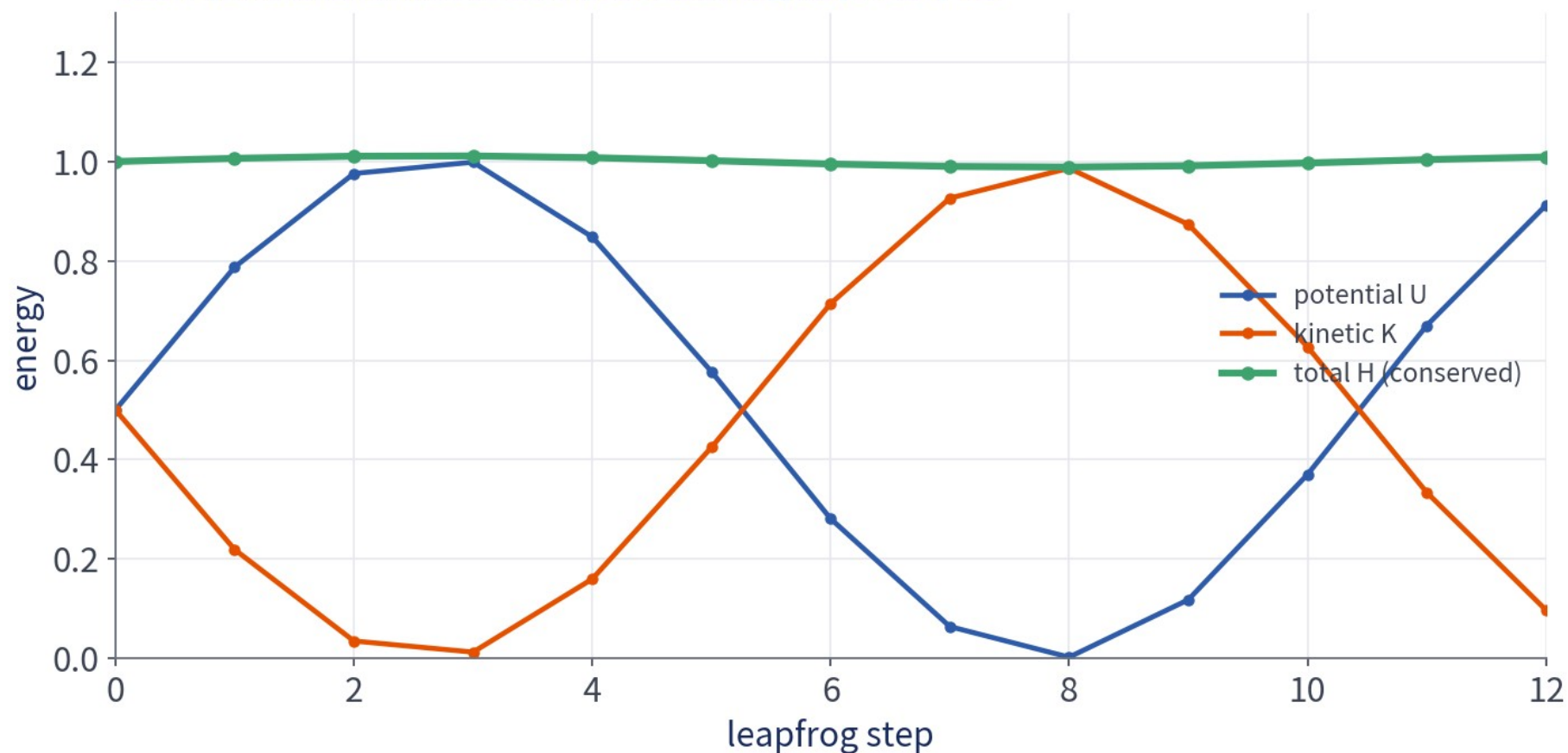
The accept step

- Accept with e-to-the-minus-that-change.
- That's 0.994 - almost always yes.
- The correction fixes leapfrog's small error.

Energy Trades, But Total Holds

As the ball rolls, potential and kinetic energy swap back and forth - the ball speeds up in the valley and slows on the rise. But their sum, the total energy, stays nearly flat across every leapfrog step. That near-conservation is why HMC accepts almost every move

Case A: potential and kinetic trade off, but total energy stays nearly flat



What Posterior Is the Ball On?

A strategy's expected return μ is a real finance posterior in disguise. Suppose a strategy's expected annual return has posterior $N(10\%, 5\%)$. Standardize it - z is the number of posterior standard deviations from the mean - and the potential becomes exactly half z -squared. The ball's position is a plausible

$$\text{posterior } \mu \sim N(10\%, 5\%), \quad z = \frac{\mu - 10\%}{5\%} : \quad z=1.0 \leftrightarrow \mu=15\%, \quad z=1.255 \leftrightarrow \mu=16.3\%$$

So the leapfrog move from $z=1.0$ to 1.255 is the sampler exploring expected returns from 15% up to 16.3%. The gradient always points back toward 10% - the posterior's most probable return.

Why the Ball Beats the Walk

Direction, not luck

The random walk

- Proposes blindly in every direction.
- Rejects most far moves.
- Crawls through the posterior.

The rolling ball

- Uses the gradient to move purposefully.
- Travels far and still gets accepted.
- Explores in a fraction of the steps.

How Far to Roll?

The one dial, and how NUTS sets it

HMC's only real dial is how long to let the ball roll. Too short, and it barely moves - a random walk again. Too long, and the ball swings out and comes right back, wasting the whole trajectory. NUTS watches the ball and stops the moment it starts turning back toward the start. We trace the swing and find that 11-turn

The trajectory

- The ball swings through the valley.
- It reaches a far point, then returns.
- Same energy, position oscillating.

What we'll do by hand

- Follow the position over the steps.
- Find where it starts moving back.
- That's where NUTS stops.

The Trajectory-Length Dial

Too short, too long, or just right

roll too short $\Rightarrow$ random walk; too long $\Rightarrow$ U-turn back (wasted); tune the length $L = \epsilon \times (\text{\#steps})$

The two failures

- Too short: tiny move, like a random walk.
- Too long: the ball U-turns back to the start.
- Both waste the gradient information.

The length

- It's the step size times the number of steps.
- Tuning it by hand is fiddly and problem-specific.
- NUTS removes the need entirely.

The Ball Swings

Position over the leapfrog steps

step	0	3	5	7	9
theta	+1.00	+1.41	+1.07	+0.36	-0.49
momentum	+1.00	-0.16	-0.92	-1.36	-1.32

The ball climbs to its farthest point near step 3, where the momentum flips negative - it has started coming back. From there it swings through the valley and out the other side. Positive position with negative momentum is the U-turn signal NUTS watches for.

By step 3 the ball has turned around: NUTS ends the roll at this U-turn.

NUTS Stops at the U-Turn

No length to tune

NUTS stops when $(\theta - \theta_0) \cdot p < 0$ (the ball turns back) $\Rightarrow$ no length to tune

The U-turn test

- Watch the displacement times the momentum.
- While positive, the ball moves away.
- When negative, it's heading back.

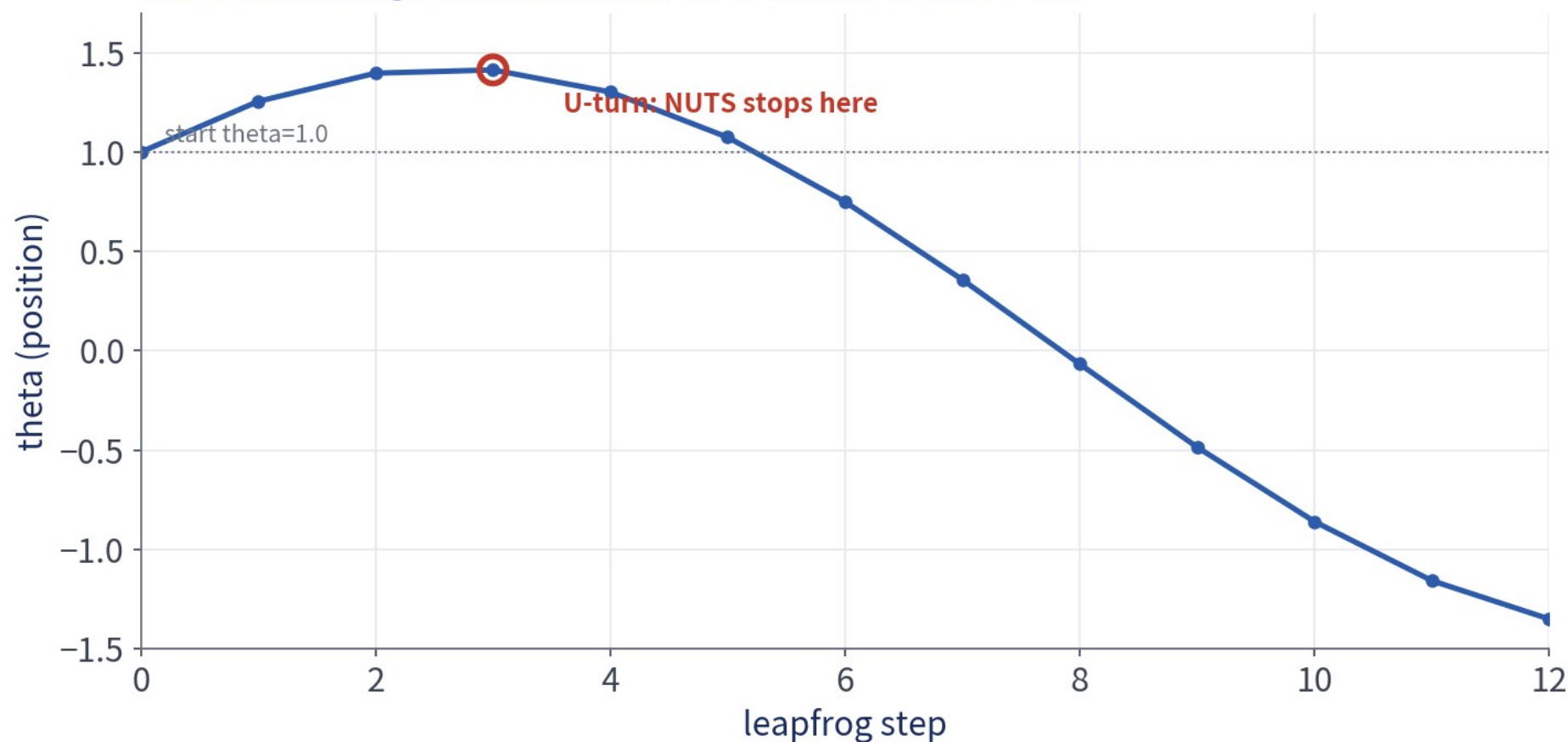
The payoff

- NUTS ends the roll right there.
- No trajectory length to set by hand.
- The longest useful move, automatically.

Out, Then Back - Stop at the Turn

The ball's position climbs to a peak near step 3, then swings back down through the start and out the far side. The circled point is the U-turn, where it first heads back toward where it began - exactly where NUTS ends the trajectory, capturing the longest useful move

Case B: the ball swings out, then turns back - NUTS ends the roll at the U-turn

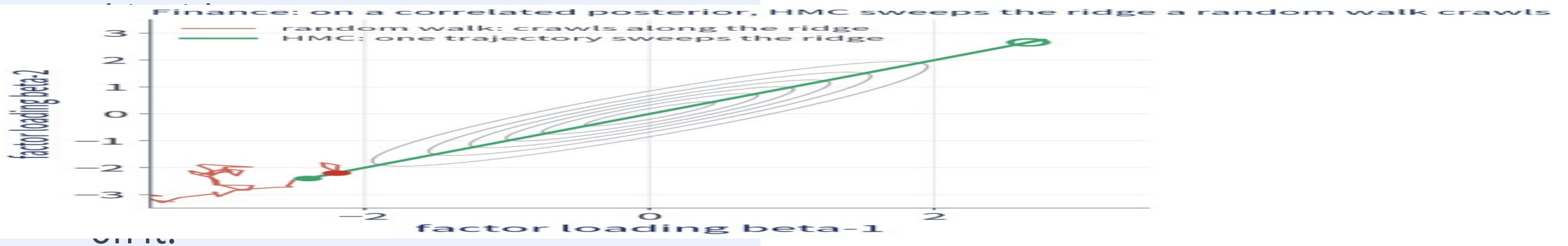


Why Finance Needs the Roll

Correlated parameters, an elongated posterior

The finance posterior

- Two overlapping factor loadings.
- Posterior correlation 0.9 - a long,



- One HMC trajectory (green) sweeps the ridge,
- then U-turns at the far end.

HMC You Don't Have to Tune

Why PyMC and Stan use NUTS

What NUTS gives you

- Automatic trajectory length, per step.
- No manual tuning of how far to roll.
- Long, efficient, accepted moves.

Why it's the default

- PyMC and Stan run NUTS under the hood.
- You write the model, it does the sampling.
- Gradient-based efficiency, no hand-tuning.

Why HMC Wins

Gradients allow moves without tuning

HMC replaces aimless wandering with directed motion. The posterior becomes a landscape, momentum sets the ball rolling, and leapfrog follows the gradient while conserving energy - so far-reaching moves are almost always accepted. NUTS then sets the one remaining dial by stopping at the U-turn. That combination is

The roll (Case A)

- Potential from the posterior, plus momentum.
- Leapfrog conserves energy.
- Acceptance near one.

The length (Case B)

- The ball swings and turns back.
- NUTS stops at the U-turn.
- No tuning, longest useful move.

Reproduce the Roll and the U-Turn

Direct Claude Code to check your arithmetic

GOAL Verify one leapfrog step, the energy conservation, and the U-turn detection.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that, for $U=0.5*\theta^2$ with gradient θ and step size 0.3:

CASE A - one step:

- * from $\theta=1$, $p=1$, do one leapfrog step;
print $\theta=1.255$, $p=0.662$, and H before/after;
- * print the acceptance probability e^{-dH} .

CASE B - the swing:

- * run 12 leapfrog steps, print θ and p each step;
- * flag the first U-turn where $(\theta-\theta_0)*p < 0$.

Then plot the energy and the position trajectory.

Match every printed number to the slides.

Check against the slides

- leapfrog: θ 1.255, p 0.662.
- H 1.000 $\rightarrow$ 1.0065.
- accept 0.994.
- U-turn near step 3.

Deliverable: `part3_practice.py` whose printed numbers match the leapfrog step and the U-turn.

Three Questions on the Cases

If you can answer these, the practice has landed

- Q1** What is the potential energy in HMC?
-
- Q2** Why is HMC's acceptance so high?
-
- Q3** How does NUTS set the trajectory length?
-

The negative log posterior - the landscape

Leapfrog nearly conserves the total energy

It stops when the ball starts turning back

All three come from the Hamiltonian roll, the energy check, and the U-turn rule.

Three Pitfalls to Avoid

Where HMC trips people

Common mistakes

- "HMC doesn't need the gradient."
- "A longer roll is always better."
- "You have to code HMC yourself."

The reality

- The gradient is essential - it steers the ball.
- Too long and the ball U-turns, wasting the move.
- PyMC and Stan run NUTS for you.

Two Calculations to Remember

Part 3 applications in one screen

1

One leapfrog step, energy conserved

From the $\theta=1$ and momentum = 1, a leapfrog step of size 0.3 moved the ball to $\theta=1.255$, and the total energy shifted only from 1.000 to 1.0065 - so the move is accepted with probability 0.994. The gradient steers, and leapfrog keeps energy nearly exact.

2

The ball swings, and NUTS stops it

Following the trajectory, the ball reached its farthest point near step 3, where the momentum flipped negative - the U-turn. NUTS ends the roll there, capturing the longest useful move with no trajectory length to tune by hand.

3

Directed motion, automatically tuned

HMC rolls a ball down the posterior using the gradient, conserving energy for high acceptance; NUTS sets the one dial by stopping at the U-turn. That is exactly what PyMC and Stan run for you.

Week 14 complete — from sampling instead of integrating, through Metropolis and Gibbs, to gradient-powered HMC and NUTS.